

Why Do We Need Random Variables?

Up to now, we have defined events as qualitative subsets of a sample space:

$$\mathcal{S} = \{\text{Heads, Tails}\} \quad \text{or} \quad \mathcal{S} = \{\text{Red, Blue, Green}\}$$

The Limitation of Qualitative Outcomes

- We cannot perform algebraic operations (like addition, averaging, or scaling) on raw outcomes like "Heads" or "Red".
- To build mathematical models, calculate averages, or measure variances, we must translate these outcomes into numerical values.

We need a systematic rule that assigns a real number to every single outcome in our sample space.

Formal Definition of a Random Variable

Definition

Random Variable Let $\mathcal{S}$ be the sample space of a random experiment. A **random variable** X is a real-valued function that maps every outcome $s \in \mathcal{S}$ to a unique real number $X(s) \in \mathbb{R}$.

$$X : \mathcal{S} \rightarrow \mathbb{R}$$

Crucial Conceptual Caveat

Despite its name, a random variable is **not a variable** and is **not random** in the traditional sense.

- It is a deterministic, mathematically defined **function**.
- The "randomness" comes solely from the underlying experiment which determines which outcome $s \in \mathcal{S}$ occurs.

The Range of a Random Variable

The collection of all numerical values that a random variable can actually take is called its **range** or **support**, often denoted by $\mathcal{T}$ or $X(\mathcal{S})$.

$$\mathcal{T} = \{x \in \mathbb{R} \mid X(s) = x \text{ for some } s \in \mathcal{S}\}$$

Example: Tossing Two Fair Coins

Let the sample space be:

$$\mathcal{S} = \{HH, HT, TH, TT\}$$

Let X be the random variable representing the **number of Heads**.

- $X(HH) = 2$
- $X(HT) = 1$
- $X(TH) = 1$
- $X(TT) = 0$

The support of X is the discrete set: $\mathcal{T} = \{0, 1, 2\}$.

Discrete vs. Continuous Random Variables

Depending on the nature of their support set $\mathcal{T}$, random variables are classified into two fundamental domains:

1. Discrete Random Variables

A random variable is **discrete** if its range $\mathcal{T}$ consists of a finite or countably infinite set of values (such as integers).

- Outcomes can be listed in a sequence: $\{x_1, x_2, x_3, \dots\}$.

2. Continuous Random Variables

A random variable is **continuous** if its range $\mathcal{T}$ consists of an uncountable interval or union of intervals of real numbers.

- It can assume any real-valued measurement in its range.

Discrete Example: Rolling Two Dice

Suppose two fair, six-sided dice are rolled, yielding a sample space $\mathcal{S}$ of 36 equally likely outcomes.

Let Y be the random variable representing the **sum of the two dice**:

$$Y(i, j) = i + j \quad \text{where } (i, j) \in \mathcal{S}$$

- The minimum possible value is: $Y(1, 1) = 2$.
- The maximum possible value is: $Y(6, 6) = 12$.
- The range is the discrete list of values:

$$\mathcal{T} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Since the range is finite, Y is a **discrete random variable**.

Continuous Example: Device Lifetime

Consider an experiment where we observe the operational lifetime (in hours) of a continuous electronic component until it experiences a structural failure.

The physical sample space consists of all non-negative real numbers:

$$\mathcal{S} = \{t \in \mathbb{R} \mid t \geq 0\}$$

Let T be the random variable representing the **lifetime of the component**:

$$T(t) = t \quad \text{for any } t \in \mathcal{S}$$

- Since the time of failure can be any fractional value (e.g., 104.532 hours), the range is the continuous interval:

$$\mathcal{T} = [0, \infty)$$

Because this range is uncountable, T is a **continuous random variable**.

Exercise: Classifying Random Variables

For each of the following scenarios, identify:

- 1 The random variable X of interest.
- 2 Whether X is **discrete** or **continuous**.
- 3 The mathematically defined range/support $\mathcal{T}$ of X .

Scenarios

- **Scenario A:** A fair coin is tossed repeatedly until the first "Heads" appears. We count the total number of tosses required.
- **Scenario B:** A researcher measures the precise blood pressure level (in mmHg) of a patient.
- **Scenario C:** A commuter records the exact waiting time (in minutes) for a city bus, which is guaranteed to arrive within 30 minutes.

The Induced Probability Map

A random variable $X : \mathcal{S} \rightarrow \mathbb{R}$ is a completely deterministic, algebraic function. It does not possess any "inherent" randomness.

Where does the probability come from?

The probability of X taking on a specific value is entirely **derived** from the probability of the underlying outcomes in the sample space $\mathcal{S}$ that map to that value.

If the underlying physical experiment is performed and yields an outcome $s \in \mathcal{S}$, the random variable evaluates to the number $X(s)$.

Translating Events to Real Numbers

Let x be a real number. We want to define what $P(X = x)$ means.

We look backward at the sample space $\mathcal{S}$ and gather all outcomes s that map to x . We call this subset the **pre-image** of x :

$$E_x = \{s \in \mathcal{S} \mid X(s) = x\}$$

Definition

The Probability Law of X The probability that the random variable X equals x is defined as the probability of its underlying event E_x occurring:

$$P(X = x) \triangleq P(\{s \in \mathcal{S} \mid X(s) = x\})$$

This is often written in compact shorthand as:

$$P(X = x) = P(s \in \mathcal{S} : X(s) = x)$$

Example: Tossing Two Fair Coins

Let $\mathcal{S} = \{HH, HT, TH, TT\}$ with equally likely outcomes ($P(s) = 1/4$). Let X represent the **number of Heads**:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

To find $P(X = 1)$, we identify which outcomes in $\mathcal{S}$ map to 1:

$$E_1 = \{s \in \mathcal{S} \mid X(s) = 1\} = \{HT, TH\}$$

Since E_1 is a standard, well-defined event in our sample space, we can calculate its probability:

$$P(X = 1) = P(E_1) = P(\{HT, TH\}) = \frac{2}{4} = 0.5$$

The number 1 is assigned the collective probability mass of the outcomes HT and TH.

Extending to Intervals and Ranges

We do not just calculate probability at single points. We often want to know the probability of a range, such as $P(a \leq X \leq b)$ or $P(X \leq x)$.

The exact same **mapping** rule applies! We collect all outcomes in the sample space $\mathcal{S}$ that map into that interval:

$$P(a \leq X \leq b) \triangleq P(\{s \in \mathcal{S} \mid a \leq X(s) \leq b\})$$

Shorthand Notation Caveat

When you write $P(X \leq x)$, you are using a convenient shorthand. The mathematical reality behind that shorthand is:

$$P(X \leq x) = P(\{s \in \mathcal{S} \mid X(s) \leq x\})$$

This translation connects the real number world ($X \leq x$) back to the set theory world ($\mathcal{S}$) where Kolmogorov's Axioms live!

Exercise 1: Cloud Database Reliability

A cloud database replicates data across 3 independent servers. The sample space $\mathcal{S}$ represents the state of each server: (x_1, x_2, x_3) where $x_i \in \{0, 1\}$ (1 = online, 0 = offline). The probability of any individual server failing (offline) is 0.1. The servers operate independently.

Let the random variable C represent the **available system throughput** (in Gbps), mapped as a function of the number of online servers

$k = x_1 + x_2 + x_3$:

$$C(s) = \begin{cases} 100 & \text{if } k = 3 \\ 50 & \text{if } k = 2 \\ 20 & \text{if } k = 1 \\ 0 & \text{if } k = 0 \end{cases}$$

Calculate the probability $P(C \geq 50)$.

Exercise 2: Reactor Pressure Telemetry

Consider an industrial pressure sensor placed in a chemical reactor. The operational pressure s (in psi) ranges uniformly over the sample space $\mathcal{S} = [0, 200]$ psi, where $P([a, b]) = \frac{b-a}{200}$ for any sub-interval.

The safety telemetry system converts the raw pressure s to a safety index Y using a piecewise logarithmic mapping:

$$Y(s) = \begin{cases} 0 & \text{if } 0 \leq s < 1 \quad (\text{noise floor}) \\ 20 \log_{10}(s) & \text{if } 1 \leq s \leq 100 \quad (\text{telemetry monitoring}) \\ 40 & \text{if } 100 < s \leq 200 \quad (\text{saturation limit}) \end{cases}$$

Your Task

- ➊ Translate the telemetry reading event $\{20 \leq Y \leq 40\}$ back to its corresponding pre-image subset $E \subseteq \mathcal{S}$.
- ➋ Calculate the probability $P(20 \leq Y \leq 40)$ using the probability law.

What is a "Distribution"?

A random variable X maps outcomes from a sample space $\mathcal{S}$ to the real numbers $\mathbb{R}$.

The way the probability mass is scattered, allocated, or "distributed" over these real numbers is called the **probability distribution** of X .

To study distributions systematically, we use specific functions:

- **The Cumulative Distribution Function (CDF)** — Applies universally to all random variables.
- **The Probability Mass Function (PMF)** — Specifically for discrete random variables.
- **The Probability Density Function (PDF)** — Specifically for continuous random variables.

Why Do We Need a Distribution Function?

Previously, calculating probabilities required looking backward to the sample space:

$$P(X \in B) = P(\{s \in \mathcal{S} \mid X(s) \in B\})$$

While mathematically sound, this approach has severe practical limitations:

- It is highly inefficient to map back to the sample space $\mathcal{S}$ for every single probability calculation.
- In many real-world applications, the original sample space $\mathcal{S}$ is complex, high-dimensional, or completely hidden from us.

The Solution

A **distribution function** acts as a mathematical proxy. Once we establish this function on the real line $\mathbb{R}$:

- **All** probabilities associated with X can be calculated using this function **alone**.

The Cumulative Distribution Function (CDF)

The most fundamental and universal way to describe the distribution of any random variable is through cumulative probabilities.

Definition

Cumulative Distribution Function Let X be a random variable. The **Cumulative Distribution Function (CDF)** of X , denoted by $F(x)$ (or $F_X(x)$), is defined for any real number x as:

$$F(x) \triangleq P(X \leq x) \quad \text{for } x \in \mathbb{R}$$

Key Concept

The value $F(x)$ represents the accumulated probability of all outcomes in the sample space that map to a value less than or equal to x .

Why $(-\infty, x]$ is Enough: Reconstructing Events

The CDF is defined strictly using semi-infinite intervals:

$$F(a) \triangleq P(X \leq x) = P(X \in (-\infty, x])$$

A Fundamental Result

Every "nice" or mathematically well-behaved subset of the real numbers can be constructed from intervals of the form $(-\infty, x]$ using basic set operations.

Consequently, the probability of any reasonable event can be written as an addition or subtraction combination of the distribution function values:

Properties of the CDF

The CDF $F(x)$ is a mathematically well-behaved function. For any random variable X , its CDF satisfies the following structural properties:

- 1 **Monotonicity:** $F(x)$ is a non-decreasing function:

$$\text{If } a \leq b, \text{ then } F(a) \leq F(b)$$

- 2 **Asymptotic Limits:**

$$\lim_{x \rightarrow \infty} F(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow -\infty} F(x) = 0$$

- 3 **Right-Continuity:** $F(x)$ is continuous from the right at every point:

$$\lim_{t \rightarrow x^+} F(t) = F(x)$$

Discrete Case: Probability Mass Function (PMF)

For discrete random variables (which take on countable values), we define probability at exact, individual points.

Definition

Probability Mass Function Let X be a discrete random variable with range $\mathcal{T}$. The **Probability Mass Function (PMF)** of X , denoted by $p(x)$ (or $p_X(x)$), is:

$$p(x) \triangleq P(X = x) \quad \text{for } x \in \mathbb{R}$$

Fundamental Axiomatic Properties

Every valid PMF must satisfy:

- $p(x) \geq 0$ for all x .
- $\sum_{x \in \mathcal{T}} p(x) = 1$.

Continuous Case: Probability Density Function (PDF)

For continuous random variables, the probability of taking on any single exact value is zero ($P(X = x) = 0$). Instead, we describe the probability over an interval using density.

Definition

Probability Density Function Let X be a continuous random variable. A non-negative, integrable function $f(x)$ is the **Probability Density Function (PDF)** of X if for any interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Axiomatic Constraints

- $f(x) \geq 0$ for all $x \in \mathbb{R}$.
- $\int_{-\infty}^{\infty} f(x) dx = F(\infty) = 1$.

Exercise: Verifying a Valid PMF

Suppose a system database query returns a variable number of matching records, X . The probability mass function of X is modeled by:

$$p(x) = \begin{cases} k \cdot x^2 & \text{if } x \in \{1, 2, 3, 4\} \\ 0 & \text{otherwise} \end{cases}$$

Your Task

- 1 Find the value of the normalization constant k such that $p(x)$ represents a mathematically valid PMF.
- 2 Determine and write down the Cumulative Distribution Function (CDF) $F(x)$ for this discrete random variable.

Exercise: Verifying a Valid PDF

Consider a continuous random variable X representing the time (in hours) a user spends actively on a platform in a session. The density is modeled by:

$$f(x) = \begin{cases} c \cdot (2x - x^2) & \text{if } x \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

Your Task

Find the value of the normalization constant c such that $f(x)$ represents a mathematically valid PDF, and determine the CDF $F(x)$.

Expectation: The Center of Mass

The **mathematical expectation** (or mean) of a random variable represents its long-run average value over infinitely repeated trials.

Physically, it acts as the **center of gravity** (centroid) of the probability distribution.

Definition

Expected Value (Discrete) Let X be a discrete random variable with support set $\mathcal{T}$ and probability mass function $p(x) = P(X = x)$. The expected value of X , denoted $E[X]$ or μ , is:

$$E[X] \triangleq \sum_{x \in \mathcal{T}} x \cdot p(x)$$

This sum is defined provided that it is absolutely convergent, i.e., $\sum |x|p(x) < \infty$.

Expectation: Continuous Analogue

For a continuous random variable, we cannot sum over individual points because any single point has zero probability.

Instead, we integrate over the continuous support using the probability density function (PDF).

Definition

Expected Value (Continuous) Let X be a continuous random variable with probability density function $f(x)$. The expected value of X is defined as:

$$E[X] \triangleq \int_{-\infty}^{\infty} x \cdot f(x) dx$$

This integral is defined provided that it is absolutely convergent, i.e., $\int |x|f(x) dx < \infty$.

Linearity of Expectation

Expectation is a **linear operator**. This means it preserves scaling and addition, regardless of whether the random variables are dependent or independent.

Theorem

Linearity Properties Let X be a random variable, and let a and b be constants. Then:

$$E[aX + b] = aE[X] + b$$

Proof: Linearity of Expectation

We prove this for the discrete case. The continuous proof follows identically by replacing summations with integrals.

Let X have support $\mathcal{T}$ and PMF $p(x)$. Apply the definition of expectation to $g(X) = aX + b$:

$$\begin{aligned} E[aX + b] &= \sum_{x \in \mathcal{T}} (ax + b) \cdot p(x) \\ &= \sum_{x \in \mathcal{T}} (ax \cdot p(x) + b \cdot p(x)) \\ &= \sum_{x \in \mathcal{T}} ax \cdot p(x) + \sum_{x \in \mathcal{T}} b \cdot p(x) \end{aligned}$$

Pulling the constants out of the summations yields:

$$E[aX + b] = a \sum_{x \in \mathcal{T}} x \cdot p(x) + b \sum_{x \in \mathcal{T}} p(x)$$

Applying $E[X] = \sum xp(x)$ and Axiom 2 ($\sum p(x) = 1$):

$$E[aX + b] = aE[X] + b \cdot (1) = aE[X] + b \quad \blacksquare$$

Variance: Quantifying Spread

While the mean tells us where the distribution is centered, it tells us nothing about how spread out the values are.

Variance measures the average squared deviation of the random variable from its mean (μ).

Definition

Variance Let X be a random variable with mean $\mu = E[X]$. The variance of X , denoted $Var(X)$ or σ^2 , is:

$$Var(X) \triangleq E[(X - \mu)^2]$$

Standard Deviation

Because variance is measured in squared units, we often use the **Standard Deviation** (σ) to keep the units consistent with the original data:

Standard Formula for Variance

Evaluating the definition $E[(X - \mu)^2]$ directly can be algebraically tedious. There exists a much simpler, standard formula.

Theorem

Computational Identity For any random variable X :

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Derivation:

Let $\mu = E[X]$. Expanding the squared term inside the expectation definition:

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2]$$

Now, apply the linearity of expectation to distribute the operator across each term:

$$\text{Var}(X) = E[X^2] - E[2\mu X] + E[\mu^2]$$

Since μ is a constant, we can pull it out of the expectation. Also, the expectation of a constant is the constant itself ($E[\mu^2] = \mu^2$):

$$\text{Var}(X) = E[X^2] - 2\mu E[X] + \mu^2$$

Substitute $E[X] = \mu$ back into the expression:

$$\text{Var}(X) = E[X^2] - 2\mu(\mu) + \mu^2 = E[X^2] - 2\mu^2 + \mu^2$$

$$\text{Var}(X) = E[X^2] - \mu^2 = E[X^2] - (E[X])^2 \quad \blacksquare$$

Properties of Variance: Scaling and Shifts

Unlike expectation, variance is **not** a linear operator. Adding a constant shift does not change the spread of the data, while scaling a variable scales its variance quadratically.

Theorem

Variance Transformations Let X be a random variable, and let a and b be constants. Then:

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Proof: Variance Transformations

Let $Y = aX + b$. First, find the mean of the transformed variable Y :

$$E[Y] = E[aX + b] = aE[X] + b$$

Now, substitute Y and $E[Y]$ into the definition of variance:

$$\begin{aligned} \text{Var}(aX + b) &= E \left[((aX + b) - E[aX + b])^2 \right] \\ &= E \left[(aX + b - (aE[X] + b))^2 \right] \end{aligned}$$

Notice that the constant shift b cancels out completely:

$$\begin{aligned} \text{Var}(aX + b) &= E \left[(aX - aE[X])^2 \right] \\ &= E \left[(a(X - E[X]))^2 \right] \\ &= E \left[a^2(X - E[X])^2 \right] \end{aligned}$$

Pulling the constant factor a^2 out of the expectation:

$$\text{Var}(aX + b) = a^2 E \left[(X - E[X])^2 \right] = a^2 \text{Var}(X) \quad \blacksquare$$

Exercise 1: Server Latency Cost Mapping

An API server experiences variable request latency. Let X represent the processing latency (in milliseconds) with the following PMF:

x	10 ms	50 ms	100 ms
$p(x)$	0.6	0.3	0.1

The operating cost (in micro-dollars) of processing a request of latency X is determined by the quadratic relationship:

$$C(X) = 2X^2 + 10$$

Your Task

Calculate the expected processing cost $E[C(X)]$ per request.

Exercise 2: Precision Sensor Noise

An industrial analog temperature sensor outputs a voltage reading V . Due to thermal interference, the sensor error $X = V - V_{\text{true}}$ (in millivolts) is a continuous random variable distributed uniformly on the range $[-5, 5]$ mV, with a constant probability density:

$$f(x) = \frac{1}{10} \quad \text{for } x \in [-5, 5]$$

Your Task

- 1 Compute the expected calibration error $E[X]$.
- 2 Compute the variance $\text{Var}(X)$ to quantify the sensor's precision limit.

Central Tendency: The Mathematical Mean

The expected value $E[X]$ we defined is physically known as the **mean** (μ). It is the most common measure of **central tendency**.

Key Characteristics of the Mean

- **Physical Centroid:** It acts as the exact balancing point of the probability mass curve or histogram.
- **Sensitivity:** Because its calculation factors in every value weighted by its probability, the mean is **highly sensitive to outliers** (skewed tails).
- **Linear Balance:** The sum (or integral) of deviations from the mean is always exactly zero:

$$E[X - \mu] = 0$$

Alternative Measures: The Median

When a distribution has extreme values or heavy asymmetry, the mean may not reflect the "typical" value. In such cases, we use the **median**.

Definition

The Median (m) The median m of a random variable X is a real number that divides the probability distribution into two equal halves.

- **Continuous Case:** m is the value satisfying:

$$F(m) = \int_{-\infty}^m f(x) dx = 0.5$$

- **Discrete Case:** m is any value satisfying:

$$P(X \leq m) \geq 0.5 \quad \text{and} \quad P(X \geq m) \geq 0.5$$

Core Advantage: The median is **robust**; changing extreme values has absolutely zero effect on its position.

Alternative Measures: The Mode

Another fundamental measure of central tendency is the **mode**, which indicates the most likely or popular outcome.

Definition

The Mode (x^*) The mode x^* is the value of x where the probability distribution reaches its maximum concentration or peak.

- **Discrete Case:** The point x^* that maximizes the PMF:

$$p(x^*) \geq p(x) \quad \text{for all } x$$

- **Continuous Case:** The point x^* that maximizes the density function $f(x)$ (typically found where $f'(x^*) = 0$ and $f''(x^*) < 0$).

Distribution Shapes

Unlike mean and median, a distribution can be **unimodal** (one peak), **bimodal** (two peaks), or **multimodal** (several peaks).

Comparing Central Tendency: Symmetry vs. Skewness

The relative positions of the Mean, Median, and Mode reveal the geometric **skewness** (asymmetry) of a distribution.

1. Perfectly Symmetric Distributions

The peaks, physical balance points, and 50% splitting lines align perfectly:

$$\text{Mean} = \text{Median} = \text{Mode}$$

2. Skewed Distributions

- **Right-Skewed (Positive):** A long right tail pulls the mean upward. The mode remains at the main peak, with the median sitting safely in between:

$$\text{Mode} < \text{Median} < \text{Mean}$$

- **Left-Skewed (Negative):** A long left tail pulls the mean downward:

$$\text{Mean} < \text{Median} < \text{Mode}$$

Robust Variability: Median Absolute Deviation

Just as the mean has a robust counterpart (the median), variance has a robust alternative known as the **Median Absolute Deviation (MAD)**. Variance and standard deviation are highly sensitive to extreme outliers because they square deviations. MAD resolves this.

Definition

Median Absolute Deviation (MAD) Let X be a random variable with median m . The Median Absolute Deviation of X is defined as:

$$MAD \triangleq \text{median}(|X - m|)$$

Why Use MAD?

- **Extreme Outlier Resistance:** A single outlier can inflate variance infinitely, whereas it barely changes the MAD.
- **Preserves Scale:** It measures variability in the same units as the original data without squaring the errors.

Exercise 3: Comparing Central Tendency Measures

Consider a continuous random variable X with the following bounded probability density function:

$$f(x) = \begin{cases} 3x^2 & \text{if } x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

Your Task

- 1 Find the expected value/mean (μ).
- 2 Calculate the median (m).
- 3 Determine the mode (x^*).
- 4 Compare these three values to identify the geometric skewness of this distribution.

Why Do We Need Joint Distributions?

In most real-world scenarios, an experiment produces multiple numerical outcomes simultaneously.

To model relationships, associations, and dependencies, we must look at variables collectively rather than in isolation.

Key Objectives

- Learn to describe the joint behavior of multiple random variables.
- Understand how to extract individual probability laws from joint laws.
- Quantify the strength and direction of linear relationships between variables.

Discrete Case: Joint Probability Mass Function

When X and Y are both discrete random variables, we describe their joint probabilities using a single function of two variables.

Definition

Joint Probability Mass Function The joint probability mass function (joint PMF) of discrete random variables X and Y is defined for all real numbers x and y as:

$$p(x, y) \triangleq P(X = x, Y = y)$$

Axiomatic Properties

Every valid joint PMF must satisfy:

- $p(x, y) \geq 0$ for all (x, y) .
- $\sum_x \sum_y p(x, y) = 1$.

Continuous Case: Joint Probability Density Function

When variables are continuous, probability at a single point is zero. Instead, we define probability over regions using a joint density surface.

Definition

Joint Probability Density Function The joint probability density function (joint PDF) of continuous random variables X and Y is a function $f(x, y)$ such that for any region $A \subseteq \mathbb{R}^2$:

$$P((X, Y) \in A) = \iint_A f(x, y) \, dx \, dy$$

Required Constraints

- $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$.
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$.

Extracting Individual Laws: Marginal Distributions

If we are given a joint distribution, we can recover the individual probability law of X or Y by summing or integrating out the other variable.

Marginal PMF (Discrete)

We sum across all possible values of the unwanted variable:

$$p_X(x) = \sum_y p(x, y) \quad \text{and} \quad p_Y(y) = \sum_x p(x, y)$$

Marginal PDF (Continuous)

We integrate across the entire domain of the unwanted variable:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Independence of Random Variables

We previously defined independence for events. Similarly, two random variables are independent if knowing the value of one provides zero information about the other.

Definition

Statistical Independence Random variables X and Y are independent if their joint distribution factors cleanly into the product of their individual marginal distributions.

- **Discrete Case:**

$$p(x, y) = p_X(x) \cdot p_Y(y) \quad \text{for all } x, y$$

- **Continuous Case:**

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y$$

Linearity of Expectation

Expectation is a **linear operator**. This means it preserves scaling and addition, regardless of whether the random variables are dependent or independent.

Theorem

Linearity Properties Let X and Y be two jointly distributed random variables. Then:

$$E[X + Y] = E[X] + E[Y]$$

Proof: Linearity of Expectation

To fully establish linearity, we must show that expectation distributes over addition for **any** two random variables X and Y , without assuming independence. Let X and Y have joint support $\mathcal{T}_{X,Y}$ and joint PMF $p(x, y)$. Apply the definition of expectation to $g(X, Y) = X + Y$:

$$\begin{aligned} E[X + Y] &= \sum_{(x,y) \in \mathcal{T}_{X,Y}} (x + y) \cdot p(x, y) \\ &= \sum_x \sum_y (x \cdot p(x, y) + y \cdot p(x, y)) \\ &= \sum_x \sum_y x \cdot p(x, y) + \sum_y \sum_x y \cdot p(x, y) \end{aligned}$$

Factoring out the variables that do not depend on the inner summation index:

$$\begin{aligned} E[X + Y] &= \sum_x x \left(\sum_y p(x, y) \right) + \sum_y y \left(\sum_x p(x, y) \right) \\ E[X + Y] &= \sum_x x \cdot p_X(x) + \sum_y y \cdot p_Y(y) = E[X] + E[Y] \quad \blacksquare \end{aligned}$$

Covariance: Quantifying Relationship Direction

While expectation tells us about the center of individual variables, we need a metric to evaluate how they vary together.

Definition

Covariance Let X and Y be random variables with means $\mu_X = E[X]$ and $\mu_Y = E[Y]$. The covariance of X and Y , denoted $\text{Cov}(X, Y)$, is:

$$\text{Cov}(X, Y) \triangleq E[(X - \mu_X)(Y - \mu_Y)]$$

Intuitive Interpretation

- **Positive Covariance:** When X is above its mean, Y tends to be above its mean (positive linear trend).
- **Negative Covariance:** When X is above its mean, Y tends to be below its mean (inverse linear trend).
- **Zero Covariance:** No linear association.

Computational Identity for Covariance

Evaluating the definition $E[(X - \mu_X)(Y - \mu_Y)]$ directly requires calculating deviations for every point, which is algebraically tedious. Instead, we use a much simpler standard identity.

Theorem

Computational Formula For any two random variables X and Y :

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Let us derive this relationship on the next slide using the properties of expectation.

Derivation: Computational Formula

Let $\mu_X = E[X]$ and $\mu_Y = E[Y]$. Expanding the terms inside the expectation definition:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\text{Cov}(X, Y) = E[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y]$$

Apply the linearity of expectation to distribute the operator across each term:

$$\text{Cov}(X, Y) = E[XY] - E[\mu_Y X] - E[\mu_X Y] + E[\mu_X \mu_Y]$$

Since the means μ_X and μ_Y are constants, we pull them out of the expectation:

$$\text{Cov}(X, Y) = E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

Substitute $E[X] = \mu_X$ and $E[Y] = \mu_Y$ back into the expression:

$$\text{Cov}(X, Y) = E[XY] - \mu_Y \mu_X - \mu_X \mu_Y + \mu_X \mu_Y$$

$$\text{Cov}(X, Y) = E[XY] - \mu_X \mu_Y = E[XY] - E[X]E[Y] \quad \blacksquare$$

Properties of Covariance

Covariance behaves according to the following mathematical rules:

Core Covariance Properties

Let X , Y , and Z be random variables, and let a and b be constants.

- ① **Symmetry:** $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- ② **Covariance with Self is Variance:** $\text{Cov}(X, X) = \text{Var}(X)$
- ③ **Scaling and Bilinearity:** $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$
- ④ **Additivity:** $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$

Variance is NOT Linear

Unlike expectation, variance is a **quadratic** operator. Because it measures squared deviations, it does not simply preserve addition or scaling.

The General Rule for Variance

Let X and Y be any two random variables, and let $a, b \in \mathbb{R}$ be constants. The variance of their linear combination is:

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

If we simply add the variables together ($a = 1, b = 1$), this reduces to the fundamental sum formula:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Key Takeaway: $\text{Var}(X + Y) \neq \text{Var}(X) + \text{Var}(Y)$ unless the variables are completely uncorrelated (i.e., $\text{Cov}(X, Y) = 0$).

Proof: Variance of a Sum

Let us formally prove that $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.
 By definition, $\text{Var}(Z) = E[(Z - E[Z])^2]$. Let $Z = X + Y$. Applying our rule of linear expectation, $E[X + Y] = E[X] + E[Y]$.

Algebraic Expansion

Group the random variables directly with their respective population means:

$$\begin{aligned}\text{Var}(X + Y) &= E \left[\left((X + Y) - (E[X] + E[Y]) \right)^2 \right] \\ &= E \left[\left((X - E[X]) + (Y - E[Y]) \right)^2 \right]\end{aligned}$$

Expand to get:

$$\begin{aligned}&= E \left[(X - E[X])^2 + 2(X - E[X])(Y - E[Y]) + (Y - E[Y])^2 \right] \\ &\implies \text{Var}(X + Y) = \text{Var}(X) + 2\text{Cov}(X, Y) + \text{Var}(Y)\end{aligned}$$

Independence and Covariance

What happens to covariance if the random variables are independent?

Theorem

Covariance of Independent Variables If X and Y are independent random variables, then:

$$\text{Cov}(X, Y) = 0$$

Proof Sketch

If X and Y are independent, then their joint expectation factors:
 $E[XY] = E[X]E[Y]$. Substituting this into our computational formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[X]E[Y] - E[X]E[Y] = 0 \quad \blacksquare$$

The Converse is not true

The converse is **not** generally true. If $\text{Cov}(X, Y) = 0$, the variables are uncorrelated, but they are **not necessarily independent**.

Exercise 1: Discrete Covariance Calculation

Let X and Y be discrete random variables representing the outcome of a simple system. Their joint PMF, $p(x, y)$, is defined by the following table:

	$Y = 1$	$Y = 2$
$X = 0$	0.1	0.4
$X = 1$	0.3	0.2

Your Task

- 1 Find the marginal PMFs of X and Y .
- 2 Compute the expected values $E[X]$ and $E[Y]$.
- 3 Compute the covariance $\text{Cov}(X, Y)$.

Exercise 2: Continuous Covariance Calculation

Let X and Y be continuous random variables. Their joint density function is defined as:

$$f(x, y) = \begin{cases} x + y & \text{if } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Your Task

- 1 Compute the expected values $E[X]$ and $E[Y]$.
- 2 Compute the expected product $E[XY]$.
- 3 Compute the covariance $\text{Cov}(X, Y)$.

The Cauchy-Schwarz Inequality

The relationship between the expectations of products and individual second moments is bounded by a fundamental mathematical law.

Theorem

Cauchy-Schwarz Inequality For any two random variables X and Y with finite variances:

$$(E[XY])^2 \leq E[X^2]E[Y^2]$$

In terms of deviations from their respective means, this is equivalent to:

$$(\text{Cov}(X, Y))^2 \leq \text{Var}(X)\text{Var}(Y)$$

Physical Implication

This inequality guarantees that covariance cannot grow infinitely large. It is bounded strictly by the individual spreads (variances) of the two variables, which allows us to define a standardized scale for relationships.

Sketch of Proof

The proof relies on the property that the expectation of a non-negative random variable is always non-negative.

1. Starting Point

For any constant $a \in \mathbb{R}$, consider the squared difference $E[(aX - bY)^2] \geq 0$.

2. Expansion

Expanding the quadratic inside the expectation:

$$E[a^2X^2 - 2aXY + Y^2] \geq 0$$

By the linearity of expectation, we obtain:

$$a^2E[X^2] - 2aE[XY] + E[Y^2] \geq 0$$

Sketch of Proof

3. The Discriminant Condition

Treating this as a quadratic in a , for the expression to be ≥ 0 for all a , the discriminant must be non-positive ($\Delta \leq 0$):

$$(2E[XY])^2 - 4E[X^2]E[Y^2] \leq 0$$

$$4(E[XY])^2 \leq 4E[X^2]E[Y^2]$$

4. Result

Dividing by 4 yields the Cauchy-Schwarz Inequality:

$$(E[XY])^2 \leq E[X^2]E[Y^2]$$

5. Equality Condition

The equality holds iff there exists an a such that $aX = Y$ wp 1.

The Correlation Coefficient: Standardizing Covariance

While covariance tells us the direction of a linear relationship, its numerical value depends on the arbitrary units of X and Y . To fix this, we normalize it.

Definition

Correlation Coefficient The correlation coefficient of two random variables X and Y , denoted by $\rho(X, Y)$ (or simply ρ), is defined as:

$$\rho(X, Y) \triangleq \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

Properties of Correlation

By direct application of the Cauchy-Schwarz inequality:

- **Boundedness:** $-1 \leq \rho(X, Y) \leq 1$.
- **Perfect Linear Relationship:** $\rho = 1$ indicates a perfect positive linear trend; $\rho = -1$ indicates a perfect negative linear trend.